

Lesson 25: Sampling & Describing Data — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Find the mean of 6, 9, 12, 13.
2. Find the median of 4, 7, 2, 9, 5.
3. Find the mode of 3, 3, 5, 8, 8, 8.
4. Find the mean of 10, 20, 30.
5. Find the median of 2, 8, 4, 6.
6. Find the range of 3, 9, 14, 5.

Section 2 — Core practice

7. Ordered data 4, 6, 9, 10, 14, 18, 21. Find Q1, Q3 and the IQR.
8. Find the IQR of 1, 3, 5, 7, 9, 11, 13, 15.
9. Name a sampling method where every k th item is chosen.
10. Name a sampling method where each member is equally likely.
11. Find the median of 12, 15, 11, 20, 9, 14.

12. Find the mean of 5, 5, 8, 10, 12.

Section 3 — Challenge & exam-style

13. Times (min), ordered: 21, 23, 26, 28, 30, 31, 34, 37, 41. Find the median and IQR.

14. A data set has mean 12 over 8 values. A 9th value of 30 is added. Find the new mean.

15. Why might a survey done only at a gym be biased about exercise habits?

16. The five-number summary is 4, 9, 15, 22, 30. State the median and the IQR.